

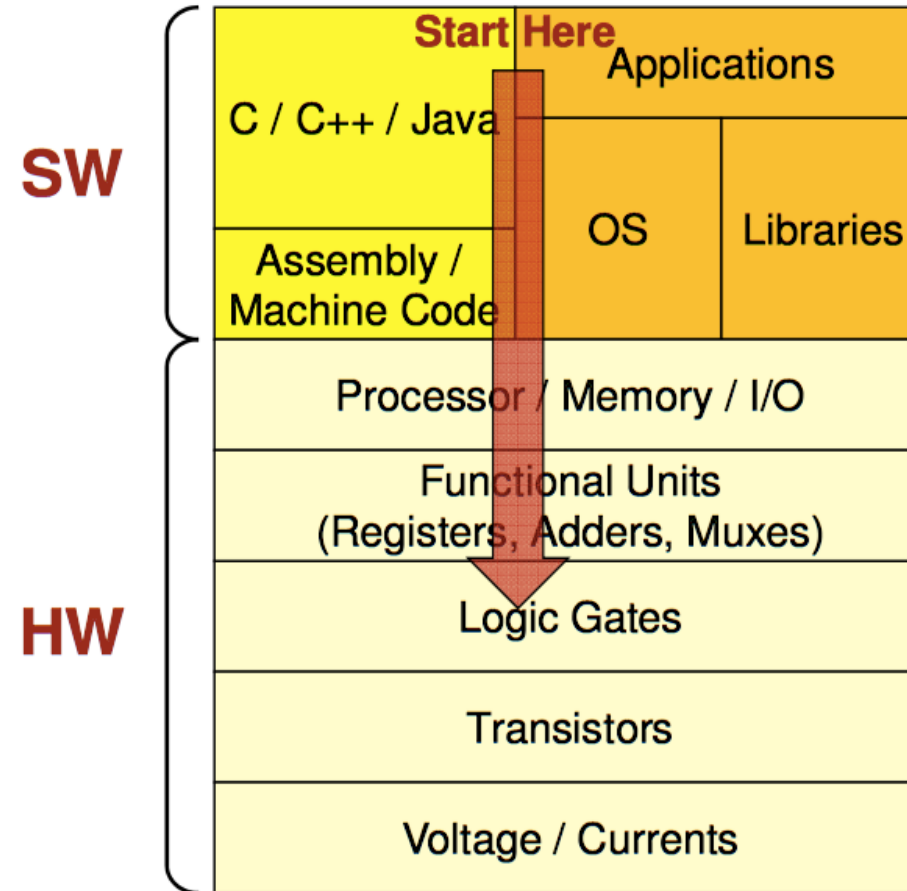
UCR



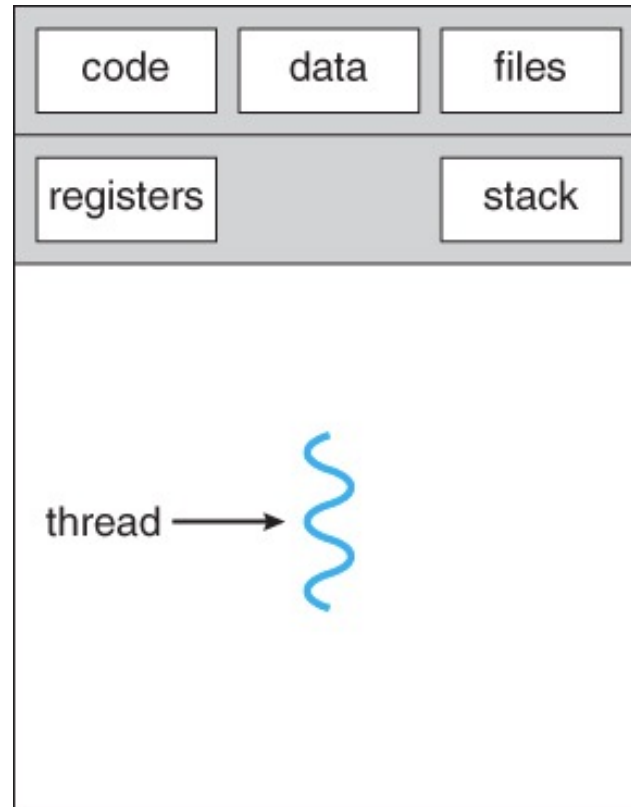
Super Quick Architecture Review

UNIVERSITY OF CALIFORNIA, RIVERSIDE

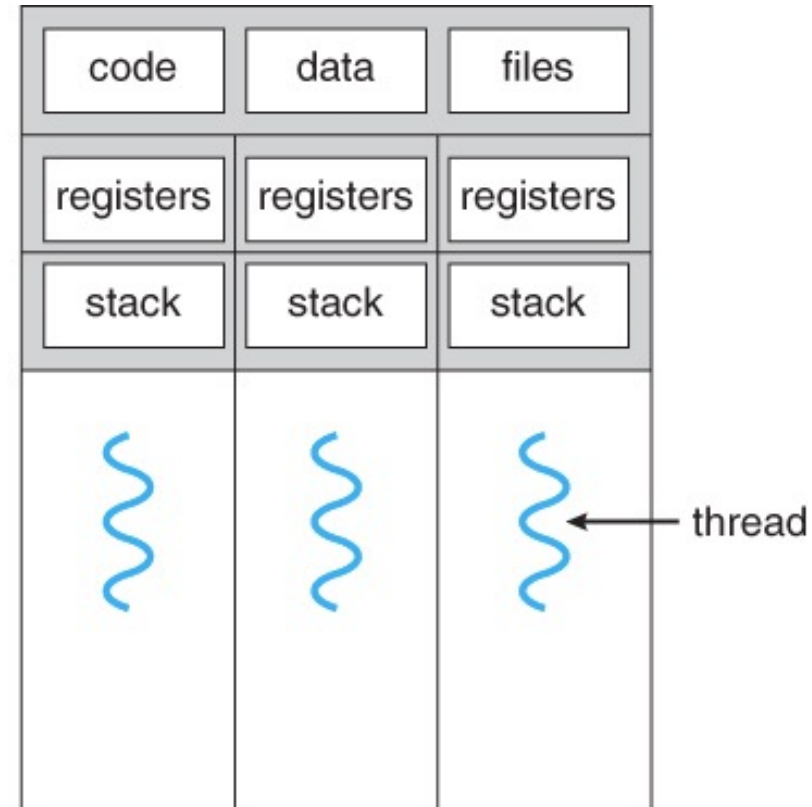
Software-Hardware stack



Process vs Thread



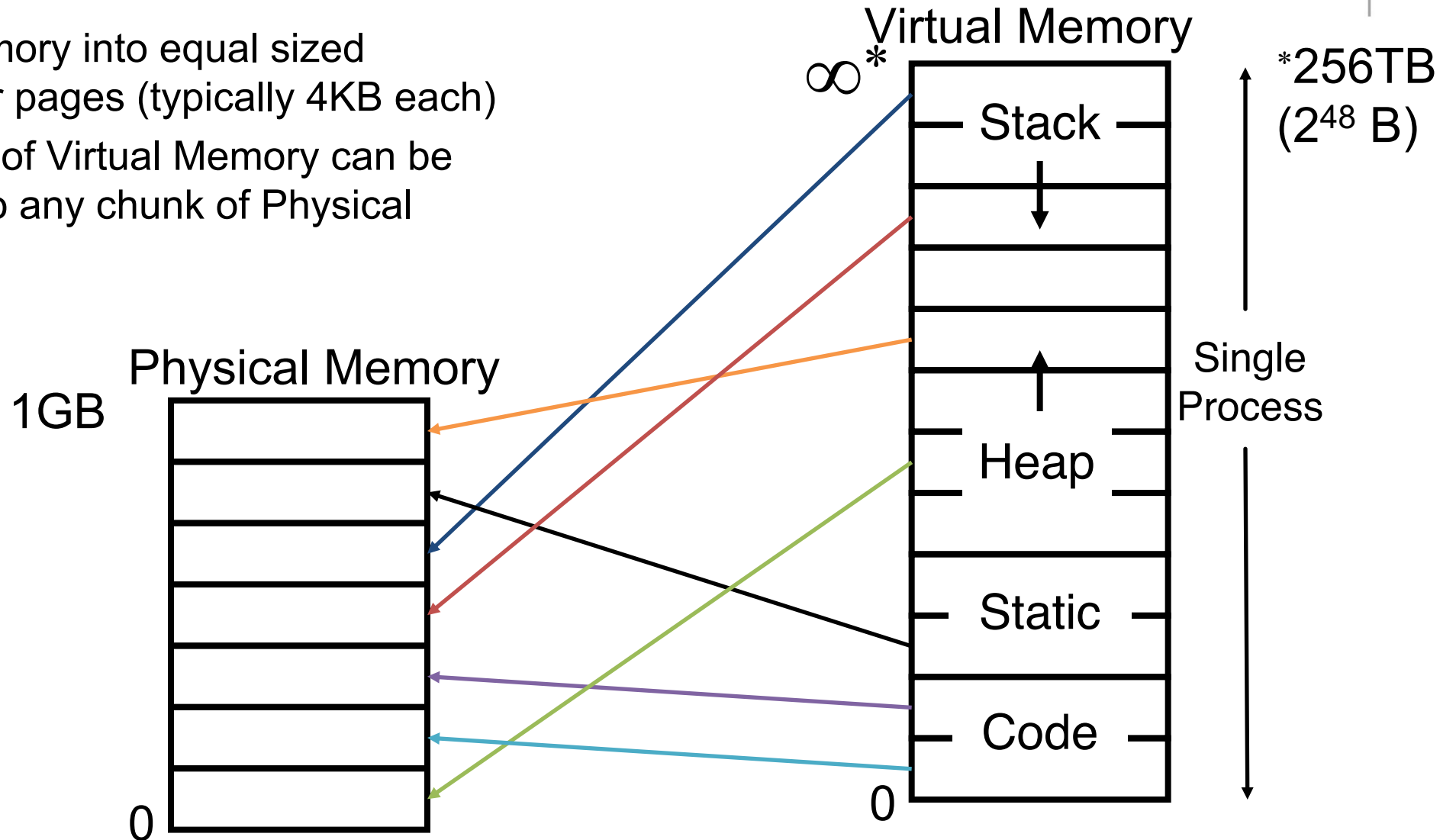
single-threaded process



multithreaded process

Virtual Memory

- › Divide memory into equal sized “chunks” or pages (typically 4KB each)
- › Any chunk of Virtual Memory can be assigned to any chunk of Physical Memory



ISA example - MIPS R-format Instructions



- Instruction fields
 - op: operation code (opcode)
 - rs: first source register number
 - rt: second source register number
 - rd: destination register number
 - shamt: shift amount (00000 for now)
 - funct: function code (extends opcode)
- Used only for ALU instructions

R-format Example



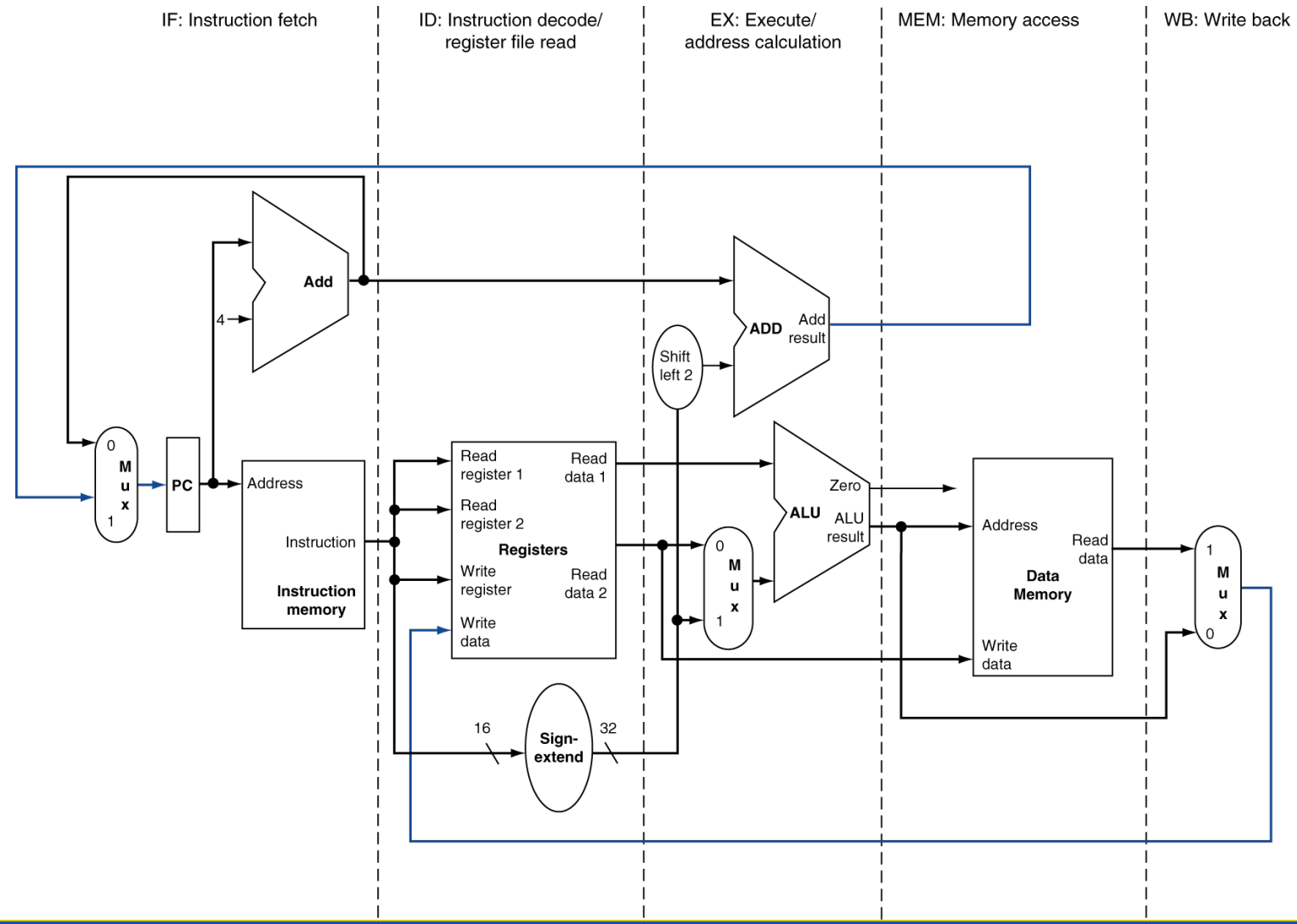
op	rs	rt	rd	shamt	funct
6 bits	5 bits	5 bits	5 bits	5 bits	6 bits

add \$r8, \$r17, \$r18

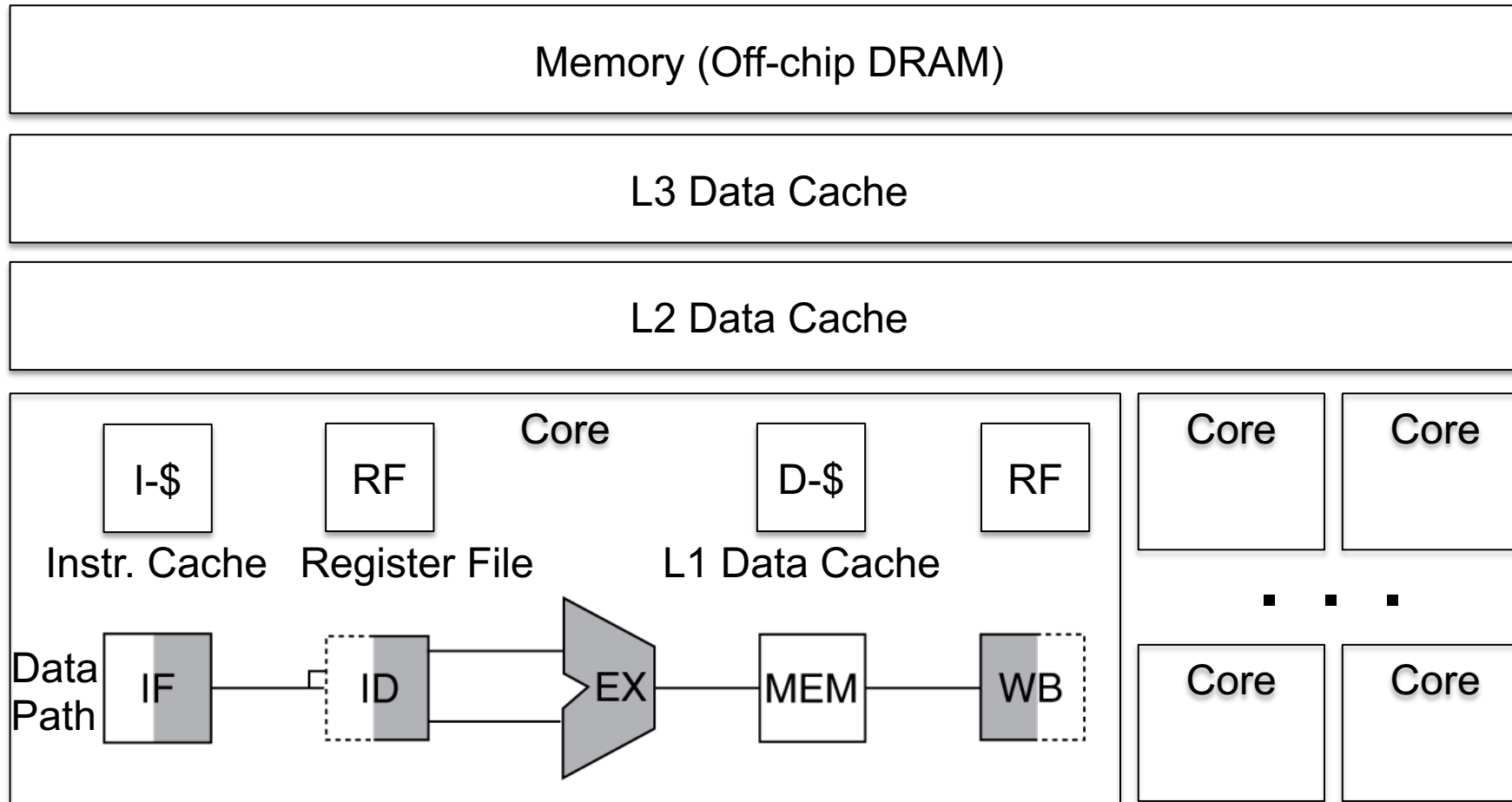
special	\$r17	\$r18	\$r8	0	add
0	17	18	8	0	32
000000	10001	10010	01000	00000	100000

$$00000010001100100100000000100000_2 = 02324020_{16}$$

5-stage pipeline



Abstracted CPU



Memory Hierarchy

